Reg. No.:

Name :

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi-110067

MID TERM EXAMINATION – MARCH 2026

Programme	**: MCA_**	**Semester**	**: SEM-II**
Course Title	**: Object Oriented Programming**	**Course Code**	**: CS-410**
Date/Session	**: 18 March 2026 / Session II**	**Slot**	**: 2:00PM-3:30PM**
Time	**: 90 Minutes.**	**Max. Marks**	**: 15**

Note: Attempt all the questions, assume suitable data if missing. Each question carry 5 marks

Q. No. **Question Description**

1 (a) Demonstrate the architecture of Java Virtual Machine (JVM) with a neat diagram. Show the roles of JRE and JDK in Java program execution.

OR

(b) Write a Java program that takes an integer as input and checks whether it is prime or not using an **if...else statement** and **a for loop**. Explain the flow of control in your program with a step-by-step execution trace.

2 (a) Write Java class named Student with attributes name, roll Number, and grade. Implement a constructor to initialize these attributes and a method displayDetails() to print student details. Create an object of the Student class and demonstrate its usage in a Java program. Explain how the concept of objects and classes is applied in this example.

OR

(b) Create a Java program demonstrating inheritance and method overriding. Define a base class Vehicle with a method move(). Then, create a subclass Car that overrides the move() method. Instantiate objects of both classes and call the move() method. Explain how inheritance and method overriding work in this example.

3 (a) How do the keywords static, final, super, and this work in Java? Provide examples for each keyword and explain their functionalities in detail.

OR

(b) Why is encapsulation important for data security?

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End Term Examination-Academic Year 205-26

Program	:	MCA	Semester	:	II
Course Title	:	Object Oriented Programming	Course Code	:	CS-410
Faculty Name	:	Subhash Chandra Patel	Date/Time	:	13-05-2026/10:00AM-1:00 PM
Time	:	3 Hrs.	Max. Marks	:	50

Question paper consists Two Sections (A and B). Suitable Options are given in each section. Each question carries equal marks

1. Explain the concept of recursion in Java with an example and write the advantages and disadvantages of using recursion Also compare recursion and iteration with suitable examples.

   OR

   Create a Java program that uses throw and throws keywords to declare and throw a custom exception. The program should simulate a banking system where an InsufficientFundsException is thrown when a withdrawal exceeds the account balance.

2. Explain the differences between iterative and recursive approaches to solving problems in Java. Provide examples of when each approach would be more suitable.

   OR

   Write a (program) class with private data members and access them using public methods. Justify your answer with discussion.

3. Examine the usage of abstract classes and interfaces. Explain the differences between them and provide examples of when each would be more suitable.

   OR

   Explain method overloading with rules and examples. Also compare method overloading and overriding.

4. Explain inheritance with suitable example also different types of inheritance and their usage.

   OR

   Create a Java application using interfaces for multiple inheritance.

5. Explain method overloading with rules and examples. Also compare method overloading and overriding.

OR

Write a Java program to demonstrate runtime polymorphism using method overriding.

Part-B

Attempt any Five

6. Design and implement a Java program that meets the requirements of the university course registration system. Your program should include:
   1. A Course class that represents a course, with attributes for course name, course number, credits, and department.
   2. A Department class that represents a department, with attributes for department name and courses.
   3. A Section class that represents a section, with attributes for section number, course, and students.
   4. A Student class that represents a student, with attributes for student name, student ID, and registered courses.
   5. Methods for registering and dropping courses, viewing course schedules, and adding, modifying, and deleting courses, departments, and sections.

7. Write a Java program that demonstrates exception handling using try, catch, and finally blocks. The program should take two numbers as input and perform division. Handle the Arithmetic Exception (division by zero) and Input Mismatch Exception (invalid input). Explain how exception handling ensures program stability in this example.
8. How does polymorphism improve the reusability and flexibility of code? Discuss with suitable example.
9. Assess the trade-offs between Encapsulation and Inheritance in Object-Oriented Programming.
10. Analyze the importance of exception handling in Java programming with suitable examples.
11. Explain the role of Java OOP concepts in Artificial Intelligence applications.

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Good Luck